An example of a positron emitter is the nitrogen atom $_7N^{13}$, which changes into the carbon atom $_6C^{13}$ by the emission of a positron, $_1e^0$ and a neutrino, i.e.:

$$_7N^{13} \rightarrow {}_6C^{13} + {}_1e^0 + {}_0\nu^0$$

III. OTHER TYPES OF SPONTANEOUS NUCLEAR TRANSMUTATION

Alpha and beta radiation are by far the most frequent types of radioactive transmutation. However, there is also a third process, already mentioned in a cursory manner, which is more or less the converse process of electron emission. In cases where it is advantageous, from the point of view of energetics, for a proton in the nucleus to change into a neutron, this transformation can be accomplished in two different ways. We have already discussed one of them—the emission of a positron. However, the same result is attained when the nucleus absorbs one of its own planetary electrons, in which case the electron, naturally, vanishes (as an absorbed photon vanishes) since its charge offsets the charge of the proton and changes the latter into a neutron. This process is called *electron capture.* As in this process no charged particle is emitted, the process can only be observed, from outside, by the consequent x-ray emission, *K-radiation,* since the absorbed electron usually comes from the innermost electron shell, the *K*-shell. As this process reduces the nuclear charge (atomic number) by one unit, the number of remaining electrons (the original number of planetary electrons minus one) is sufficient for the newly formed atom. But the remaining planetary electrons must now re-group themselves; an electron originally occupying a position in shell farther from the nucleus falls into and occupies the place vacated by the captured electron. This produces the x-ray emission, which is the *K*-radiation characteristic of the element in question. At the same time, in order to conserve the angular momentum or spin value of the atom—since the captured electron had a spin value of 1/2 h—a neutrino is emitted.

Such a process occurs, for instance, in the beryllium nucleus $_4Be^7$, an